CIV: Content Integrity Verification for Distributed Intrusion Detection Systems*

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Abstract

In distributed IoT environments, an intrusion detection system (IDS) typically aggregates reports from individual nodes. However, prior work largely assumes that such reports are trustworthy and has not sufficiently addressed the risk that compromised nodes can manipulate their reports—without altering the underlying network traffic—to distort the IDS input. To address these limitations, we propose the Content Integrity Verification (CIV) framework. CIV leverages gateway-level observations as an independent reference and quantitatively measures the standardized discrepancy between each node's report and the corresponding observation. CIV then excludes reports exceeding a predefined threshold from aggregation, thereby mitigating the impact of manipulated inputs on IDS performance. Experimental results show that the average F1-score across all evaluation settings was 0.266 without defense, whereas it improved to 0.704 with CIV. In addition, CIV outperformed existing Byzantine-robust aggregation methods in 48 out of 60 baseline-comparison $\alpha-\beta$ settings (80%) and achieved an average F1-score approximately 31% higher than Trimmed Mean, the strongest baseline.

1 Introduction

In IoT environments, distributed intrusion detection system (IDS) architectures are widely adopted, in which multiple nodes independently monitor network traffic, and an upper-layer edge server aggregates their monitoring results for intrusion detection [1]–[3]. This architecture is motivated by the difficulty of transmitting raw traffic directly to a central server under bandwidth constraints, latency requirements, and privacy concerns [3], [4]. Accordingly, practical deployments commonly rely on node-level reports, where each device summarizes its local observations and the resulting reports are aggregated as input to an upper-layer IDS [3].

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However, while this distributed architecture improves scalability and efficiency, it also introduces an important vulnerability that has received relatively limited attention. Existing distributed IDS pipelines implicitly assume that reports generated by individual nodes are trustworthy. When a node is compromised, however, an attacker can manipulate the feature-level report delivered to the upper layer without modifying the underlying network traffic. Such report-level falsification distorts the IDS input representation and can substantially degrade detection performance, even when attack traffic is present in the network.

Prior work on robustness in distributed systems has primarily focused on mitigating extreme outliers through statistical aggregation methods, such as Byzantine-robust averaging [5]. However, these approaches rely only on the mutual consistency of reports across nodes and do not leverage an independent external reference. Consequently, these methods may be ineffective when an attacker crafts manipulated reports to resemble normal patterns. As a result, verifying whether each node's report faithfully reflects the underlying network traffic—i.e., report integrity—remains an insufficiently explored gap in existing distributed IDS research.

To address these limitations, this study proposes the Content Integrity Verification (CIV) framework for distributed IDS. The key idea of the proposed approach is to use gateway-level observations, independently collected at network bottlenecks, as a reference for verification. CIV quantitatively measures the discrepancy between each node's report and the corresponding gateway observation, and mitigates the impact of manipulated inputs on the IDS by suppressing the contribution of suspicious reports to aggregation.

In particular, CIV does not attempt to recover corrupted information; rather, it serves as a defensive mechanism that preserves attack-related signals in honest reports by preventing manipulated reports from dominating the aggregated output. This approach also imposes a structural trade-off on the attacker. The more strongly an attacker falsifies reports toward benign patterns to evade IDS detection, the larger the discrepancy from gateway observations becomes, thereby increasing the likelihood of detection by CIV. Experimental results show that, under honest-majority conditions, CIV restores IDS performance to a level close to the clean baseline and outperforms four representative Byzantine-robust baselines in 48 out of 60 baseline-comparison settings.

The main contributions of this work are as follows:

- We propose the CIV framework, which uses gateway observations as a verification reference to evaluate report integrity and mitigate the impact of manipulated inputs.
- We show that CIV improves IDS performance by mitigating input contamination rather than recovering corrupted information, and systematically evaluate its effectiveness across different falsification strengths (α) and node compromise ratios (β).
- We identify the operational boundary of CIV by showing that performance degradation under high node compromise ratios is not a CIV-specific failure, but is instead associated with a broader structural limitation of aggregation-based defenses.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the system and threat model. Section 4 presents the proposed method. Section 5 introduces the experimental setup. Section 6 provides performance evaluation and analysis. Section 7 discusses limitations and open issues. Finally, Section 8 concludes the paper.

2 Related work

This section reviews prior work directly related to CIV. It covers ML/DL-based IDSs and distributed IDSs in IoT environments, hierarchical deployment strategies for IDS operations, Byzantine-robust aggregation, and studies on data integrity and trustworthiness in distributed systems. These studies provide important contributions to detection performance, cooperative system design, deployment optimization, robust aggregation, and trustworthiness assurance. Nevertheless, they generally do not sufficiently address the challenge of directly verifying report integrity before aggregation using an independent reference, namely, whether each node's report faithfully reflects the underlying observations.

2.1 ML/DL-Based and Distributed IDS in IoT Environments

In IoT environments, IDS techniques have shifted from conventional rule-based and signature-based methods toward machine learning (ML)- and deep learning (DL)-based approaches [6]–[9]. Conventional ML methods have achieved strong detection performance using predefined network-level features, whereas DL models, such as CNNs and RNNs, have been used to improve detection accuracy while reducing reliance on manual feature engineering [7], [8]. Accordingly, existing IDS research has primarily been designed and evaluated around performance metrics such as detection accuracy and false positive rate [7], [9].

To address the scalability and bandwidth limitations of centralized IDSs, distributed and cooperative IDS architectures have been actively studied in IoT environments [10]. In these architectures, each node monitors local traffic, while integrated analysis is performed by an upper-layer entity or through collaboration among nodes [10], [11]. This line of work has developed along two main directions: reducing communication overhead and energy consumption through selective reporting and lightweight feature extraction, and improving detection performance by combining information across nodes to exploit spatial and temporal correlations [10], [12]. Furthermore, distributed IDS research has explored various cooperative mechanisms, including alert correlation, cooperative model learning, and consensus-based updates [11]–[13].

However, these studies generally assume that input data collected in centralized or semi-centralized environments faithfully reflects the underlying network state [7], [8]. Distributed IDS research similarly assumes that reports submitted by individual nodes are trustworthy inputs. Furthermore, trust-based cooperative IDSs primarily focus on long-term reputation-based trust estimation and do not adequately address real-time integrity verification at the per-report level [12].

2.2 Computational Placement in IoT IDS

In addition to detection performance, IDS placement across network layers has been considered an important design issue in IoT environments [10], [14]. Prior work has examined various deployment architectures across layers, ranging from IoT devices and gateways to fog/edge nodes and the cloud [14], [15]. In particular, gateways and IoT devices face limitations in performing computationally intensive ML/DL-based detection locally due to CPU, memory, and energy constraints [14], [15]. Accordingly, many studies have adopted architectures that offload detection tasks to fog or edge servers, while fog-based distributed IDSs and edge-assisted federated learning approaches have primarily addressed performance-oriented objectives such as latency, resource constraints, and communication efficiency [16], [17].

Meanwhile, the gateway layer is important for IDS placement and the design of verification mechanisms because gateways, despite their limited computational resources, can be positioned at points through which network traffic must pass. However, prior IDS research has primarily considered

this positional advantage from the perspective of performance optimization, rather than fully leveraging it for report integrity verification.

2.3 Byzantine-Robust Aggregation

The problem of robustly aggregating inputs in the presence of adversarial participants has been extensively studied in distributed machine learning and federated learning (FL) for Byzantine fault tolerance [18]. For example, Krum selects the update with the smallest sum of distances to its closest updates [19]. Coordinate-wise Trimmed Mean and Median mitigate the influence of outliers along each coordinate and provide statistically robust aggregation rules [20]. These approaches have also been extended to IIoT environments through geometric-median-based variants [5], [21].

However, these methods share a common limitation: they identify outliers based solely on statistical relationships among the submitted inputs. In other words, because they lack an independent external reference and rely only on relative differences among inputs, they may fail to distinguish manipulated inputs that mimic benign patterns [19], [20]. Indeed, Xie et al. and Fang et al. showed that such robust aggregation rules can be circumvented or their performance can be substantially degraded through inner-product manipulation attacks and optimization-based local model poisoning, respectively [22], [23].

These limitations also apply to IDS environments. When a compromised node submits traffic reports that appear to fall within the benign range while misrepresenting actual observations, aggregation based solely on inter-node consistency may fail to effectively detect such manipulation. Therefore, beyond conventional robust aggregation, a report integrity verification mechanism based on an independent reference is needed to determine whether each node's report reflects actual observations.

2.4 Data Integrity and Trust in Distributed Systems

Data integrity and trustworthiness in distributed systems have been studied through various approaches, including secure aggregation, blockchain-based immutability, and device-level trust scoring [24]–[29]. These studies make important contributions to ensuring integrity and security during data transmission, storage, aggregation, and model training. However, most approaches focus on post-generation stages and do not directly verify report-level integrity—namely, how accurately each node's report reflects actual observations.

For example, the secure aggregation protocol by Bonawitz et al. preserves privacy by aggregating data without revealing individual contributions, but it does not verify the correctness of the contributions themselves [24]. In addition, HBFL by Sarhan et al. uses blockchain to support the integrity and traceability of model updates, but it does not consider the trustworthiness of the original IDS inputs from which those updates are generated [29]. Furthermore, studies on IoT trust management and FL-based IDS address long-term trust estimation and the trustworthiness of training data, respectively, but they do not provide a mechanism for directly verifying report integrity, i.e., the consistency between individual reports and actual observations, before aggregation [26]–[28].

2.5 Research Gap and Positioning

The studies reviewed above have contributed to the development of distributed IoT IDS architectures from different perspectives. ML/DL-based IDS research has focused on improving detection accuracy and reducing false positive rates, while distributed IDS research has aimed to improve scalability and communication efficiency through cooperative architectures.

In addition, computation placement studies have addressed the deployment of detection tasks under resource constraints, while Byzantine-robust aggregation and distributed trust studies have proposed methods to improve aggregation robustness and system trustworthiness under adversarial conditions.

However, these studies share a common limitation: they do not directly verify, prior to aggregation, whether the reports submitted by individual nodes faithfully reflect actual network observations. In other words, although prior work has broadly addressed the performance, scalability, and aggregation robustness of distributed IoT IDSs, it has not explicitly formulated report integrity itself as an independent verification target. This gap is particularly important because attackers can distort the IDS input representation by manipulating only the reports transmitted to higher layers, even without directly altering the underlying network traffic.

To address this gap, CIV leverages gateway-level observations as an independent reference and quantitatively evaluates the discrepancy between each node's report and the corresponding gateway observation prior to aggregation. By doing so, CIV mitigates the impact of manipulated inputs before they affect the upper-layer IDS and provides a new approach for systematically addressing report integrity verification in distributed IDS environments.

3 System and Threat Model

3.1 System Overview and Model

This study assumes a distributed intrusion detection system (IDS) environment in which multiple IoT nodes monitor traffic, and detection is performed at an edge server. Each IoT node monitors its local traffic and generates a node-level report $r_i(t)$, which summarizes activity within a fixed time window using a common set of flow-level features. The generated reports are transmitted to the edge server, where they are aggregated and used as input to the IDS.

Meanwhile, gateways are deployed at network choke points and independently observe traffic through flow-based network monitoring. Accordingly, the gateway observation $o_i(t)$ is defined as an observation vector directly derived from the underlying network traffic and does not depend on information reported by individual nodes. In other words, the IDS pipeline operates on node-level reports, whereas the gateway collects traffic information through an observation path independent of the nodes.

In this study, $r_i(t)$ and $o_i(t)$are defined over the same fixed time window. Here, $r_i(t)$ denotes the report generated by node i by summarizing its local traffic, whereas $o_i(t)$ denotes the gateway observation obtained by aggregating traffic flows observed during the same time interval in a device-centric manner. Specifically, $o_i(t)$ reflects both inbound and outbound traffic flows attributable to device i within the time window at time t. As a result, the device representation includes not only traffic generated by the device itself, but also observations of externally originated attack traffic received by the device.

Both representations consist of flow-level statistics, such as byte counts, packet counts, duration, and connection state, and the gateway observations rely only on L3/L4-level metadata without payload inspection. Therefore, the proposed gateway-based observation model is applicable even in environments with encrypted traffic. Furthermore, because $r_i(t)$ and $o_i(t)$ are defined in the same feature space, the two representations can be directly compared.

3.2 Threat Model

In this study, we assume that a fraction $\beta \in [0,1]$of IoT nodes may be compromised. Compromised nodes can distort the input to the upper-layer IDS by falsifying locally generated node-level reports $r_i(t)$before transmission. This setting follows an adversarial framework that is structurally related to Byzantine distributed detection models, where compromised sensors transmit fictitious observations to a fusion center [30], [31].

The attacker attempts to evade detection by crafting falsified reports to resemble benign patterns. For a compromised node, the falsified report is defined as follows.

$$\tilde{r}_i(t) = (1 - \alpha)x_i(t) + \alpha\mu_i^{ben} \quad (1)$$

Here, $x_i(t)$ denotes the original flow-level report vector before falsification, and μ_i^{ben} denotes the benign reference value of node i estimated during the reference estimation interval. In addition, $\alpha \in [0,1]$ is a parameter that controls the falsification strength. As α increases, the falsified report moves closer to the benign reference value; $\alpha = 0$ corresponds to the original report without falsification, whereas $\alpha = 1$ corresponds to complete replacement by the benign reference value.

This attack model builds on the abstract falsification channel assumed in prior Byzantine literature and instantiates it as a benign-reference-directed falsification strategy suitable for IoT IDS environments [30], [31]. This implies that the attacker falsifies reports toward the statistical center of benign behavior.

The attack targets flow-level reports transmitted to the upper layer rather than the underlying network traffic itself. Consequently, an attacker can distort the IDS input representation without modifying network packets. In particular, when manipulated reports are crafted to mimic a benign distribution, existing aggregation-based defenses that rely on inter-node consistency can be effectively bypassed.

We assume that gateway observations $o_i(t)$ are generated by independently monitoring network traffic and are not directly affected by node-side report manipulation. Furthermore, we assume that the gateway itself and the gateway–edge communication channel are trustworthy or, at a minimum, integrity-protected. This assumption is required for using gateway observations as an independent external verification reference. This trust boundary is necessary because CIV uses gateway observations as an external verification reference. If the gateway is compromised or if gateway-to-edge observations are tampered with, the external reference itself may become unreliable, limiting the effectiveness of CIV under such scenarios.

This study does not consider traffic-level attacks, such as packet injection and packet suppression, or gateway compromise. These attacks correspond to threat models that are fundamentally different from report falsification and may be addressed as separate research topics.

4 Proposed Method

This study proposes the Content Integrity Verification (CIV) framework. CIV identifies suspicious reports by quantitatively evaluating the discrepancy between each node's report and the corresponding gateway-level observation. This verification process is performed prior to data aggregation and mitigates the impact of input contamination on the upper-layer IDS by suppressing the contributions of suspicious reports. Rather than restoring corrupted information, CIV operates as a report integrity verification mechanism that maintains trustworthy inputs through filtering. The overall procedure consists of the following steps.

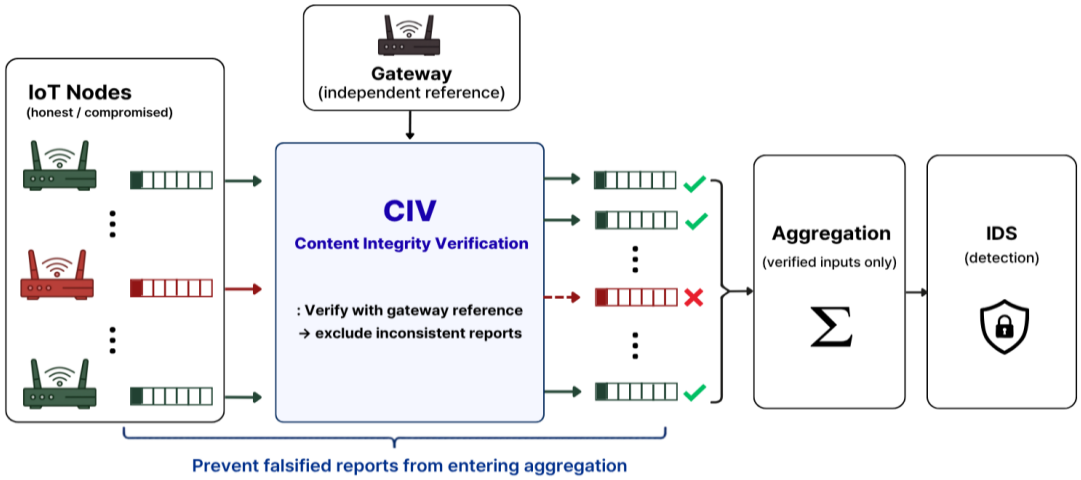


Figure 1: Proposed CIV framework

4.1 Benign Reference and Standardized Discrepancy

In the first step of CIV, reference statistics representing benign behavior are estimated for each node. Specifically, using gateway-level observations collected during the reference estimation interval, CIV computes the mean $\mu_{i,k}^{ben}$ and standard deviation $\sigma_{i,k}^{ben}$ for each feature dimension k. Here, the superscript ben indicates that the corresponding statistic is estimated during the reference estimation interval. Based on these statistics, the discrepancy between node i's report $r_i(t)$ and the corresponding gateway observation $o_i(t)$ at time t is defined as follows.

$$\Delta_i(t) = \sum_{k=1}^{d} \left(\frac{r_{i,k}(t) - o_{i,k}(t)}{\sigma_{i,k}^{ben}} \right)^2 \tag{2}$$

Here, i denotes the node index, t denotes the time-window index, d denotes the total number of feature dimensions, and k denotes the feature-dimension index. In addition, $r_{i,k}(t)$ and $o_{i,k}(t)$ denote the k-th feature values of node i's report and the corresponding gateway observation at time t, respectively.

4.2 Thresholding and Hard Exclusion

The computed discrepancy score $\Delta_i(t)$ is used as a criterion for determining whether each report falls within the benign range. In this study, the threshold τ is estimated from the benign discrepancy distribution obtained during the reference estimation interval. Specifically, τ is defined as the p-th percentile of the benign discrepancy distribution, as follows.

$$\tau = Q_p\left(\{\Delta_i(t) \mid (i,t) \in \mathcal{D}_{\text{cal}}\}\right) \tag{3}$$

Here, $Q_p(\cdot)$ denotes the p-th percentile, and $\mathcal{D}_{cal}$ is the set of discrepancies observed under honest reporting. In this study, we use $p = 95$. This corresponds to a one-sided criterion that treats deviations in the upper 5% of the discrepancy distribution under benign conditions as anomalous.

We then apply the following hard exclusion strategy to reports that exceed this threshold.

$$w_i(t) = \begin{cases} 1, & \text{if } \Delta_i(t) \le \tau \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

This rule includes a report in aggregation only when its discrepancy falls within the benign range; otherwise, the report is treated as a likely manipulated input, and its contribution is completely removed. This threshold-based binary removal rule is designed to account for the sensitivity of aggregation-based IDSs to even small residual contamination. Accordingly, we adopt deterministic removal rather than soft weighting to suppress the influence of manipulated reports before aggregation. However, under high-β settings, this design may reduce the number of valid reports and trigger fallback. The resulting fallback behavior is defined in §4.3, and its impact is analyzed in §6.4.

4.3 CIV-based Aggregation

Using the computed weights $w_i(t)$, the aggregated report vector $\hat{r}(t)$at time t is defined as follows.

$$\hat{r}(t) = \frac{\sum_i w_i\,(t)\, r_i(t)}{\sum_i w_i\,(t)} \tag{5}$$

This equation indicates that the aggregate is constructed from individual reports verified as consistent with gateway-level observations. The denominator $\sum_i w_i\,(t)$ normalizes the aggregate so that its scale remains consistent despite changes in the number of valid reports.

However, if all reports exceed the threshold in a given window t, resulting in $\sum_i w_i\,(t) = 0$, the above equation is undefined. In such cases, we replace the aggregated representation for that window with the unweighted mean of all reports and define this operation as a fallback. This fallback mechanism preserves an aggregated representation even when no valid reports are available. This fallback behavior is analyzed in §6.4.

5 Experimental Setup

5.1 Dataset and Preprocessing

This study used the TON_IoT dataset, which provides flow-level network traffic. Because the dataset contains flow-level information observable at the gateway level, it is suitable for experimentally implementing the proposed system model.

To emulate a distributed IoT monitoring environment, traffic records were organized into outbound and inbound views using source and destination IP information, respectively, and aggregated into device-centric node representations over fixed time windows. This process produced node-level time-series inputs for each node i at each time window t and provided a controlled logical node emulation for evaluating report falsification in distributed IDS settings.

Furthermore, to implement the system model described in §3.1, we constructed node-level reports $r_i(t)$ and gateway-level observations $o_i(t)$ from traffic within the same time intervals using aligned flow-level feature representations.

From the full TON_IoT dataset, we selected five traffic classes: normal, DDoS, DoS, scanning, and backdoor. This selection was not intended to comprehensively cover diverse attack types, but rather to preserve feature-space comparability between $r_i(t)$ and $o_i(t)$ after aggregation.

5.2 IDS Model

For intrusion detection, this study used a Random Forest (RF) classifier. The model is trained on clean aggregated data, and its detection performance is evaluated at inference time under three input conditions: clean input, falsified input, and CIV-corrected input. Because this study focuses on analyzing the impact of report falsification and CIV on aggregation-based IDSs rather than comparing classifier architectures, a single IDS model was used throughout the experiments.

5.3 Attack Configuration

In this study, we experimentally instantiate the report-level threat model defined in §3.2. At each time window t, we randomly select a subset of IoT nodes according to the compromise ratio β and falsify the original report vector $x_i(t)$ of each selected node as follows.

$$\tilde{r}_i(t) = (1-\alpha)x_i(t) + \alpha\mu_i^{ben} \quad (1)$$

Here, $\alpha \in [0,1]$ denotes the falsification strength, and β denotes the compromise ratio, i.e., the fraction of compromised nodes among all IoT nodes. We evaluated the attack settings over parameter combinations of $\alpha \in \{0.1,0.2,\ldots,1.0\}$ and $\beta \in \{0.1,0.2,\ldots,0.9\}$, resulting in 90 non-degenerate attack settings. Some figures visualize representative subsets of the full evaluation grid for readability.

For clarity, we divide the evaluation settings into two regimes. The primary evaluation regime corresponds to $\beta \leq 0.5$, representing a typical operational setting in which at least half of the nodes remain honest. In contrast, the stress regime corresponds to $\beta \geq 0.6$, representing a highly compromised setting in which aggregation-based verification and mitigation become difficult.

The case $\beta = 1.0$ was excluded from the evaluation because it corresponds to a degenerate case in which all nodes are compromised. In this case, no trustworthy set of reports remains available for aggregation, making it difficult to meaningfully define the notion of performance recovery through aggregation-based mitigation. Therefore, our experiments focus on performance degradation under partial-compromise settings and the mitigation effect of CIV.

5.4 Evaluation Metrics

In this study, we used F1-score, TPR, FPR, and AUROC to evaluate IDS performance. F1-score was used as the primary metric for comparing changes in overall classification performance, TPR was used to assess the attenuation of attack-related signals, and FPR was used to examine whether false positives increased after applying CIV. AUROC was used to evaluate the classifier's overall discriminative ability independently of a specific threshold setting. Based on these metrics, IDS performance was compared across three conditions: clean input, falsified input, and CIV-corrected input.

Additionally, ΔF_1, defined as the absolute improvement in F1-score under CIV-corrected input relative to falsified input, was used as a supplementary metric. We also included an honest-only reference, constructed solely from reports submitted by uncompromised nodes, for comparison. This reference was computed by applying the same aggregation procedure after assigning a weight of 0 to

manipulated reports and a weight of 1 to honest reports; it serves as a reference for the performance achievable when compromised reports are perfectly excluded under partial-compromise settings. When no honest report remains available in a given window, the honest-only reference follows the same fallback rule defined in §4.3.

6 Performance evaluation and analysis

The experiments were conducted on 23 IoT nodes from the TON_IoT dataset over 90 attack settings, consisting of $\alpha \in \{0.1, 0.2, \ldots, 1.0\}$ and $\beta \in \{0.1, 0.2, \ldots, 0.9\}$. For the no-defense and CIV recovery analyses, we report results over the full attack grid. For the baseline comparison, we focus on a representative subset of 60 α–β settings used in Fig. 5 to compare CIV with four Byzantine-robust aggregation baselines. The results are presented in the following order: the impact of report falsification (§6.1), the performance recovery effect of CIV (§6.2), comparison with baselines (§6.3), analysis of the operational boundary (§6.4), ablation study (§6.5), and computational and storage overhead analysis (§6.6).

6.1 Impact of Report Falsification

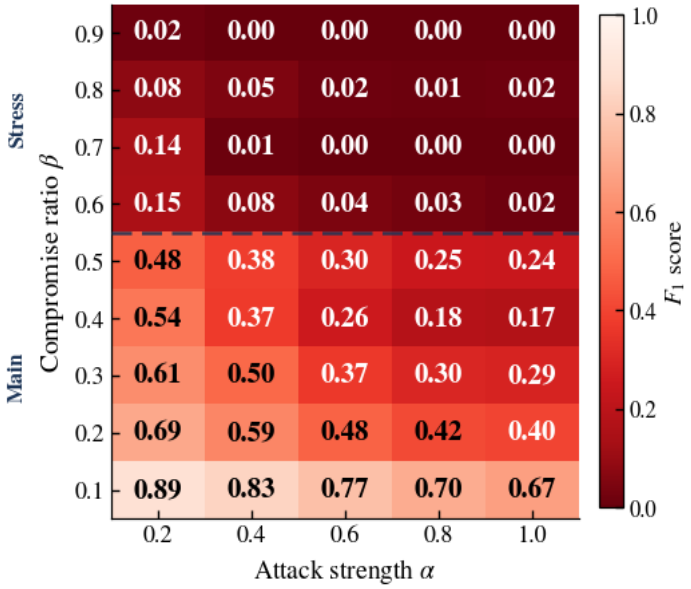


Figure 2: IDS F1 under report falsification (no defense)

Report falsification under the attacker model defined in this study can substantially degrade IDS performance without altering the underlying network traffic itself. As shown in Fig. 2, the IDS F1-score remains high at 0.979 under the clean setting, whereas performance generally decreases as the compromise ratio β and falsification strength α increase after falsification is introduced. For example, at $\beta = 0.4$ and $\alpha = 1.0$, the F1-score drops to 0.17; for $\beta > 0.7$, it falls to 0.05 or lower under most conditions.

These results indicate that β, the node compromise ratio, and α, the falsification strength, jointly affect IDS performance. Under low-β settings, the impact of falsified reports is partially mitigated through aggregation; however, as the proportion of compromised nodes increases, performance

degradation caused by increasing α becomes more pronounced. This occurs because, when an attacker falsifies reports toward the benign reference value, the manipulated reports bias the aggregate toward the benign distribution. As a result, attack-related signals in honest reports are diluted, and the IDS input representation moves closer to the benign region relative to the decision boundary.

Furthermore, these results empirically demonstrate that the report-level falsification threat model defined in §3 can cause substantial performance degradation. This provides a baseline for interpreting the performance recovery achieved by CIV in the following sections and motivates the need for report integrity verification using gateway-level observations.

6.2 Recovery with CIV

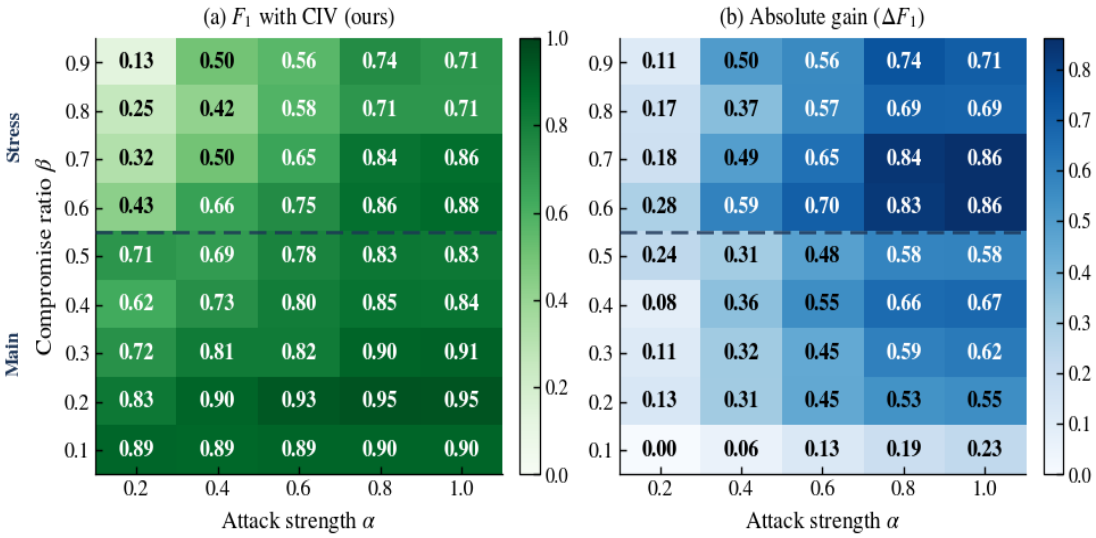


Figure 3: CIV recovery across compromise and attack regimes

Figure 3 shows the F1-score after applying CIV, along with the absolute performance improvement ΔF_1 relative to falsified input, summarized over a representative subset of α–β conditions. Overall, CIV demonstrates strong performance recovery under low compromise-ratio regimes. In particular, for $\beta \leq 0.3$, performance is restored to a level close to the clean baseline in most cases, and even at $\beta = 0.4$, substantial performance improvement is observed overall. For example, under the condition $\beta = 0.4$ and $\alpha = 1.0$, the no-defense F1-score drops to 0.17, whereas it recovers to 0.84 after applying CIV. This case illustrates a representative example of performance recovery under a single condition, and, on average across all 90 evaluation settings, the F1-score improves substantially from 0.266 to 0.704.

In contrast, performance recovery becomes increasingly limited under high compromise-ratio regimes. At $\beta = 0.6$, CIV maintains relatively high performance, whereas at $\beta = 0.8$, the F1-score remains in the range of 0.25–0.71. This behavior is not primarily due to CIV's detection failure, but is instead associated with the structural effect of increasingly frequent fallback conditions as the number of valid reports decreases. A detailed analysis is provided in §6.4.

In addition, as α increases, performance under the no-defense condition decreases, whereas ΔF_1 generally increases. This is because stronger falsification toward the benign reference direction makes the IDS more susceptible, while simultaneously increasing the discrepancy with gateway-level observations, thereby improving the detectability of falsified reports by CIV.

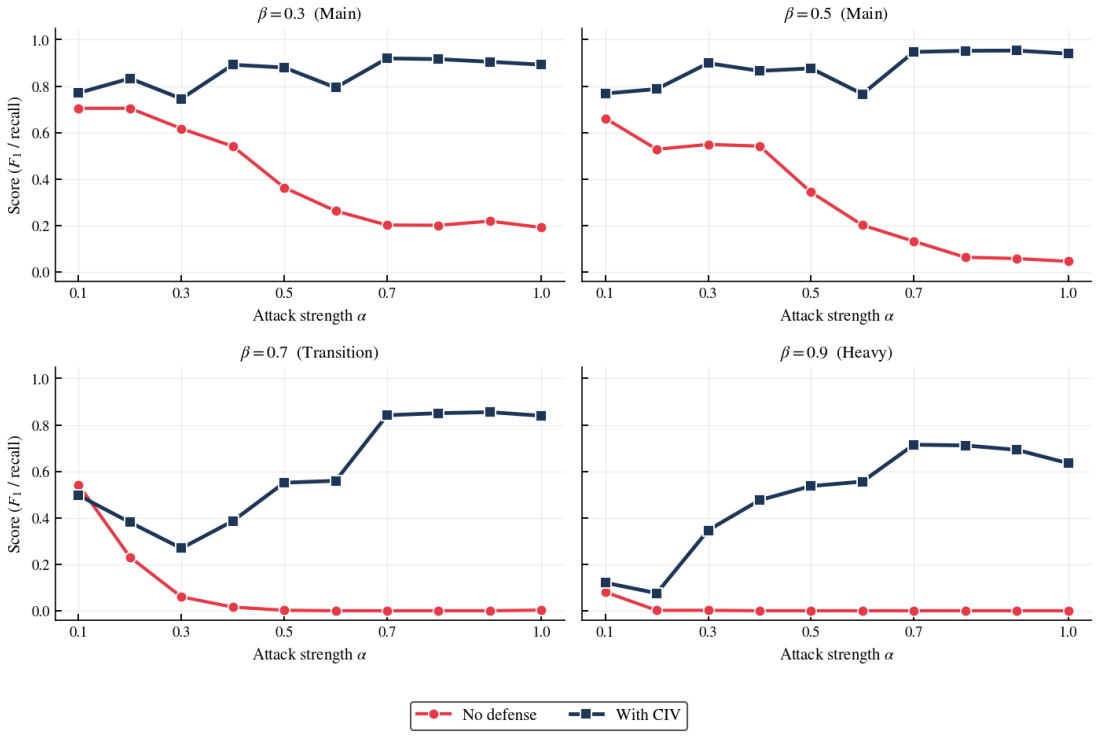


Figure 4: Attacker's dilemma, stronger falsification harms IDS but improves CIV detectability

Figure 4 shows the performance trends of the no-defense condition and the CIV-applied condition as the falsification strength α increases. Overall, as α increases, the F1-score under the no-defense condition decreases, whereas the performance after applying CIV remains relatively stable under low compromise-ratio regimes. In particular, for $\beta = 0.3$ and $\beta = 0.5$, CIV maintains high performance even as the no-defense performance consistently degrades.

These results indicate that as an attacker falsifies reports more strongly toward the benign reference, the IDS becomes more susceptible, while the discrepancy with gateway-level observations increases, thereby strengthening the CIV verification signal. In other words, as the falsification strength increases, both the potential for IDS evasion and the CIV-based discrepancy signal increase simultaneously, indicating the presence of an inherent structural trade-off.

Specifically, in the regime $\beta \leq 0.3$, CIV maintains performance close to the clean baseline under most conditions. At $\beta = 0.5$, overall performance remains high, although degradation is observed in some cases. At $\beta = 0.7$, a moderate level of recovery is still achieved, whereas at $\beta = 0.9$, performance preservation becomes limited. This degradation is not primarily due to CIV detection failures, but rather to the structural limitations of the aggregation process, where the number of valid reports decreases, and fallback becomes more frequent as the compromise ratio increases. A quantitative analysis of this effect is provided in §6.4.

6.3 Comparison with Byzantine-Robust Baselines

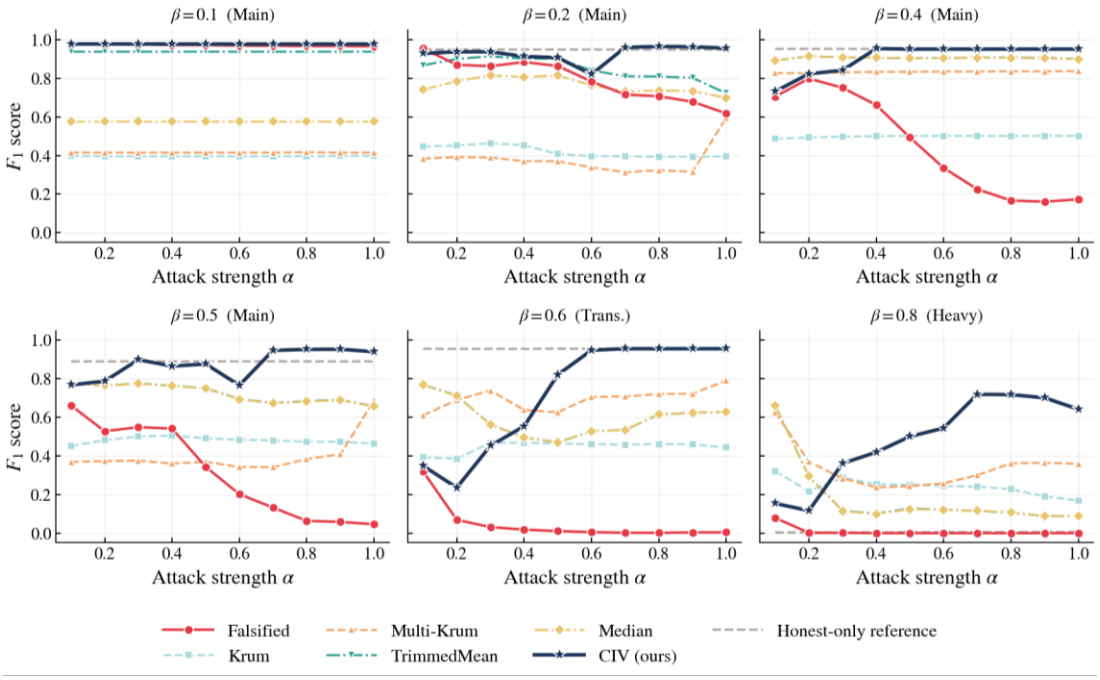


Figure 5: Comparison with Byzantine-robust aggregation baselines

Figure 5 compares the proposed CIV with representative Byzantine-robust aggregation methods and an honest-only reference. The baselines include Krum, Multi-Krum, coordinate-wise Trimmed Mean, and Median. The honest-only reference is computed by applying the same aggregation procedure after assigning a weight of 0 to manipulated reports and a weight of 1 to honest reports based on ground-truth compromise labels. For readability, baseline comparisons are visualized over representative compromise regimes rather than the full evaluation grid.

Experimental results show that CIV achieves higher F1-scores than all baselines in 48 out of 60 baseline-comparison α–β settings (80%). This performance advantage becomes more pronounced as the compromise ratio increases, whereas in low-compromise regimes such as $\beta = 0.1$, where the impact of falsification is limited, most methods—including CIV—maintain relatively high performance.

Across the 60 baseline-comparison settings, CIV attains an F1-score of 0.704, outperforming the strongest baseline, Trimmed Mean (F1 = 0.536), by an absolute margin of 0.168 points (approximately 31% relative improvement). These results indicate that CIV not only performs well under specific conditions but also provides consistent performance gains across the baseline-comparison settings.

In particular, the effectiveness of CIV becomes more pronounced in moderate-compromise regimes. At $\beta = 0.4$, CIV achieves F1-scores in the range of 0.624–0.846, and under $\alpha \geq 0.5$, it attains 0.793–0.846, approaching the average level of the honest-only reference (0.826). Moreover, under $\beta = 0.6$ and $\alpha \geq 0.5$, CIV maintains F1-scores in the range of 0.747–0.878, whereas the best-performing baseline reaches only 0.717.

In contrast, the existing baselines exhibit different forms of performance limitations. Krum shows consistently low performance, while Multi-Krum, Trimmed Mean, and Median maintain reasonable performance under low-compromise regimes but degrade significantly when $\beta \geq 0.6$. This suggests

that conventional aggregation methods, which rely solely on relative consistency among node reports, fail to effectively distinguish manipulated inputs that mimic the benign distribution.

In summary, CIV leverages gateway-level observations as an external reference for report integrity verification, thereby providing more stable and consistent defensive performance than existing Byzantine-robust aggregation methods across most conditions.

6.4 Operational Scope and Fallback Analysis

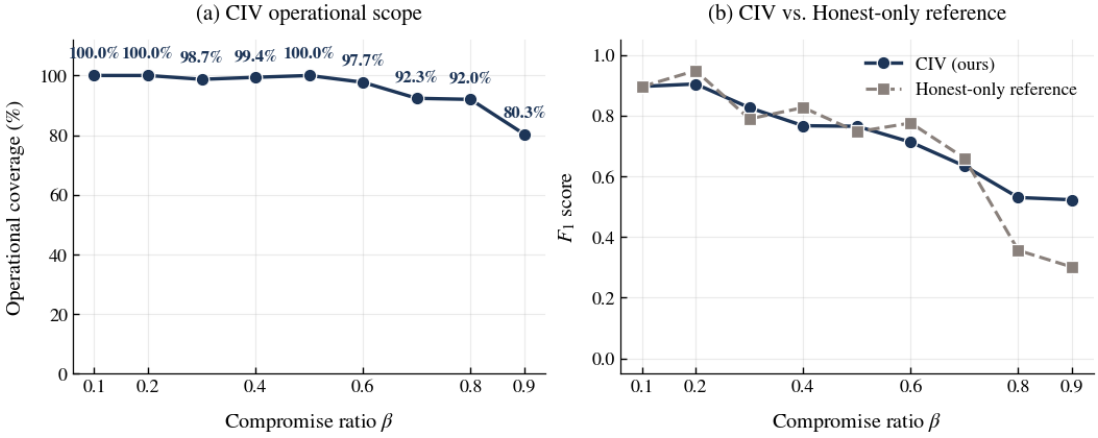


Figure 6: Operational Scope of CIV across compromise regimes

Figure 6 illustrates the operational scope of CIV under different compromise regimes. The hard exclusion mechanism of CIV falls back to the unweighted mean when all reports exceed the threshold; in this study, we define operational coverage as the proportion of time windows in which CIV operates without triggering fallback.

As shown in Fig. 6(a), the operational coverage remains close to 100% for $\beta \leq 0.5$, and still achieves a high level of 92.0% at $\beta = 0.8$. In contrast, at $\beta = 0.9$, the coverage decreases to 80.3%, indicating a substantial increase in fallback-triggering windows under the extreme compromise regime. This indicates that as the compromise ratio increases, the number of windows triggering fallback gradually rises, potentially limiting the operational range in which passive CIV can function reliably.

Fig. 6(b) compares CIV with the honest-only reference. For $\beta \leq 0.6$, the two methods show similar performance. However, in high-compromise regimes where $\beta \geq 0.7$, CIV achieves higher F1-scores than the honest-only reference. This suggests that, under extreme compromise conditions, even ideally removing compromised reports may leave too few honest reports to reliably preserve sufficient IDS signals. Therefore, the honest-only reference should be interpreted as a passive-verification reference rather than an absolute upper bound under structurally under-supported aggregation conditions.

In summary, CIV maintains high operational coverage and meaningful defensive effectiveness for $\beta \leq 0.8$. However, under the extreme compromise condition of $\beta = 0.9$, operational coverage decreases substantially, indicating that stable defense may be limited when relying solely on passive filtering of already-received reports. Therefore, additional verification mechanisms, such as active confirmation, may be required in such environments.

6.5 Design Justification and Ablation Study

This section provides empirical justification for the main design components of CIV presented in §4: the discrepancy measure, hard exclusion, and threshold estimation method. To this end, we conducted an ablation study for each component under the same experimental setting.

First, comparison of discrepancy measures shows that the feature-wise standardized discrepancy score (§4.1) provides the most consistent performance. The proposed method achieves a higher average F1-score overall than the Mahalanobis-based variant, and it also shows clearer weight separation between honest and compromised nodes. This suggests that covariance estimation can become unstable in high-dimensional flow-level feature spaces, empirically supporting the design choice in §4.1.

We then compare hard exclusion and soft weighting. The results show that their average performance difference is not large, but hard exclusion provides more consistent results overall. In particular, under high-compromise regimes, hard exclusion more effectively suppresses the residual influence of manipulated reports, supporting the validity of the binary exclusion design adopted in §4.2. Finally, Fig. 7 presents the sensitivity analysis results for the threshold percentile p.

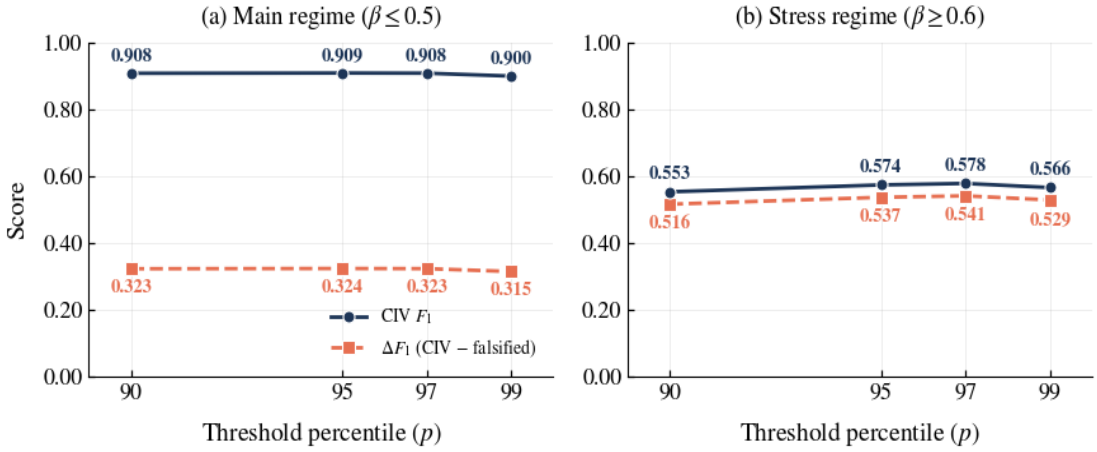


Figure 7: Sensitivity of CIV performance to threshold percentile p ∈ {90, 95, 97, 99}.

As shown in Fig. 7, the performance variation across different threshold percentiles is limited within the range of $p \in \{90,95,97,99\}$. Specifically, in the main regime ($\beta \leq 0.5$), the F1-score remains between 0.900 and 0.909, with a variation of only 0.009. In the stress regime ($\beta \geq 0.6$), the F1-score ranges from 0.553 to 0.578, corresponding to a variation of 0.025. These results indicate that the overall performance is not highly sensitive to the choice of threshold percentile.

The relative performance gain of CIV, measured by ΔF_1, is larger in the stress regime, ranging from 0.516 to 0.529, compared to the main regime. This indicates that performance degradation due to falsified input is more pronounced under high-compromise conditions, and that CIV provides substantial improvement by mitigating the impact of input contamination.

In addition, both $p = 95$ and $p = 97$ show stable performance across the main regime ($\beta \leq 0.5$) and the stress regime ($\beta \geq 0.6$). Based on these results, we select $p = 95$ as the default setting in this study.

In conclusion, the ablation results demonstrate that the key design choices of CIV are empirically well-founded. In particular, feature-wise standardized discrepancy and hard exclusion show consistent performance in suppressing input contamination, and the threshold percentile remains relatively robust within a reasonable range.

6.6 Computational and Storage Overhead

Metric	CIV
Mean latency per window	5.174ms
95th percentile latency per window	5.608ms
Additional verification cache	5.75KB

Table 1: Runtime latency and additional storage overhead of CIV

To evaluate the practical overhead of CIV, we measured its latency and additional verification cache under a representative setting of α=0.8 and β=0.5. The measurement was conducted on 1,620 test windows using 5 warm-up runs followed by 100 measured repetitions. As shown in Table 1, CIV required 5.174 ms per window on average and 5.608 ms at the 95th percentile. CIV also required only 5.75 KB of additional verification cache for storing the reference information used in report–observation discrepancy verification. These results show that the additional verification cost remains at the millisecond level with limited storage overhead in the evaluated environment. However, its feasibility on resource-constrained edge or gateway hardware requires further implementation-based validation.

7 Discussion and open issues

This study demonstrates that the CIV framework can effectively address report falsification in IoT environments; however, several important limitations remain regarding its applicability and experimental setup.

7.1 Operational scope and gateway trust assumptions.

CIV assumes that a sufficient number of valid reports remain after filtering in each aggregation window to sustain aggregation. As observed in §6.4, operational coverage remains high up to $\beta = 0.8$, but decreases to 80.3% at $\beta = 0.9$ under extreme-compromise conditions, where fallback becomes increasingly frequent. This suggests that the performance degradation in this regime is not primarily due to a specific design failure, but rather reflects a structural limitation in which aggregation itself becomes difficult to sustain reliably. Therefore, under extreme-compromise conditions, CIV alone may not provide sufficient defense, and additional sources of integrity evidence or complementary verification mechanisms are required.

In addition, CIV assumes that gateway observations remain independent of node-side report manipulation and that the gateway-to-edge reporting channel is integrity-protected. If the gateway itself is compromised or the gateway-to-edge reporting channel is tampered with, the verification reference may become unreliable. Such scenarios fall outside the current threat model and may require complementary protection mechanisms.

7.2 Dataset and IDS model specificity.

The experiments in this study are limited to the TON_IoT network-flow subset and a Random Forest-based IDS. Although the core concept of CIV is not directly tied to a specific IDS architecture, the effectiveness of aggregation-based defense may vary depending on the classifier's decision

boundary and its sensitivity to the input distribution. For example, in anomaly-based IDSs or deep learning-based classifiers, the same level of contamination suppression may lead to different changes in detection performance. Therefore, the generalizability of CIV across diverse IDS families and datasets requires separate systematic validation. In addition, the current evaluation assumes that gateway observations are sufficiently aligned with node-level reports and provide a reliable external reference. In real deployments, measurement noise, partial visibility, or routing asymmetry may distort report–observation alignment and affect discrepancy estimation; evaluating CIV under noisy or partially observable gateway settings remains future work. Furthermore, the current logical-node emulation does not explicitly capture multiple gateways, node mobility, NAT-induced device identity ambiguity, or heterogeneous device behavior; validating CIV under these deployment factors remains future work.

7.3 Static threshold calibration.

In this study, the threshold τ is estimated once from benign data in the reference estimation interval and then used as a fixed value. This design may be bypassed if an attacker adjusts the falsification strength below the threshold or gradually induces drift in the reference statistics over time, thereby weakening the verification criterion itself. While the proposed attack model suggests an inherent trade-off between reducing the report–observation discrepancy and maintaining IDS-evasion effectiveness, adaptive adversarial strategies designed to remain below the threshold are not systematically evaluated in the current study.

In addition, under extreme-compromise conditions, filtering alone may be insufficient, and additional integrity evidence or complementary verification mechanisms may be required.

8 Conclusion

This study showed that, in distributed IDS environments, node-level report falsification can distort the input representation of aggregation-based IDSs and substantially degrade detection performance, even without directly modifying the underlying network traffic. To mitigate this problem, this study proposed the Content Integrity Verification (CIV) framework, which uses gateway-level observations as an independent reference. CIV is distinguished from existing Byzantine-robust aggregation methods in that it does not attempt to recover corrupted information or modify the detection model itself; rather, it provides a report integrity verification-based defense that preserves attack-related signals contained in honest reports by suppressing the influence of manipulated reports at the input stage.

Experiments on the TON_IoT dataset showed that CIV achieved strong performance recovery under low- and moderate-compromise conditions. CIV recorded higher F1-scores than four representative Byzantine-robust baselines in 48 out of 60 baseline-comparison α–β settings (80.0%). In addition, we observed a structural trade-off: as the falsification strength increases, the potential for IDS evasion increases, but the discrepancy from gateway-level observations becomes larger, thereby increasing the likelihood of detection by CIV.

At the same time, this study clearly identifies the operational boundary of CIV. In high-compromise regimes, the honest-only reference also shows performance degradation, suggesting that the limitation in this regime stems not from the failure of a specific method, but from the structural reduction in the number of valid reports available for aggregation. Nevertheless, even under these conditions, CIV maintains superior performance compared with existing Byzantine-robust aggregation baselines and outperforms the honest-only reference in some high-compromise regimes.

In summary, this study empirically demonstrates that report integrity verification should be treated as an independent research problem in distributed IDS environments. The results further suggest that

gateway-level observations can serve as an effective external reference for evaluating whether node-level reports faithfully reflect underlying network activity. Future work will investigate integration with active verification, robustness under more realistic noise conditions, and generalizability across diverse IDS models and deployment settings. More broadly, CIV contributes to trustworthy distributed monitoring in IoT-enabled digital infrastructure by improving the reliability of security information used in edge-assisted IDS operations.

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References

[1] Jullian, O., Otero, B., Rodriguez, E., Gutierrez, N., Antona, H., & Canal, R. (2023). Deep-learning based detection for cyber-attacks in IoT networks: A distributed attack detection framework. *Journal of Network and Systems Management, 31*(2), 33. https://doi.org/10.1007/s10922-023-09722-7

[2] Bhavsar, M. H., Bekele, Y. B., Roy, K., Kelly, J. C., & Limbrick, D. (2024). FL-IDS: Federated learning-based intrusion detection system using edge devices for transportation IoT. IEEE Access, 12, 52215–52226. https://doi.org/10.1109/ACCESS.2024.3386631

[3] Rashid, M. M., Khan, S. U., Eusufzai, F., Redwan, M. A., Sabuj, S. R., & Elsharief, M. (2023). A federated learning-based approach for improving intrusion detection in industrial Internet of Things networks. Network, 3(1), 158–179. https://doi.org/10.3390/network3010008

[4] Tariq, U., Ahmed, I., Bashir, A. K., & Shaukat, K. (2023). A critical cybersecurity analysis and future research directions for the Internet of Things: A comprehensive review. Sensors, 23(8), 4117. https://doi.org/10.3390/s23084117

[5] Li, S., Ngai, E. C.-H., & Voigt, T. (2024). An experimental study of Byzantine-robust aggregation schemes in federated learning. IEEE Transactions on Big Data, 10(6), 975–988. https://doi.org/10.1109/TBDATA.2023.3237397

[6] Khraisat, A., & Alazab, A. (2021). A critical review of intrusion detection systems in the Internet of Things: Techniques, deployment strategy, validation strategy, attacks, public datasets and challenges. Cybersecurity, 4(1), 18. https://doi.org/10.1186/s42400-021-00077-7

[7] Al-Garadi, M. A., Mohamed, A., Al-Ali, A. K., Du, X., Ali, I., & Guizani, M. (2020). A survey of machine and deep learning methods for Internet of Things (IoT) security. IEEE Communications Surveys & Tutorials, 22(3), 1646–1685. https://doi.org/10.1109/COMST.2020.2988293

[8] Jamalipour, A., & Murali, S. (2022). A taxonomy of machine-learning-based intrusion detection systems for the Internet of Things: A survey. IEEE Internet of Things Journal, 9(12), 9444–9466. https://doi.org/10.1109/JIOT.2021.3126811

[9] Jayalaxmi, P. L. S., Saha, R., Kumar, G., Conti, M., & Kim, T.-H. (2022). Machine and deep learning solutions for intrusion detection and prevention in IoTs: A survey. IEEE Access, 10, 121173–121192. https://doi.org/10.1109/ACCESS.2022.3220622

[10] Arisdakessian, S., Wahab, O. A., Mourad, A., Otrok, H., & Guizani, M. (2023). A survey on IoT intrusion detection: Federated learning, game theory, social psychology, and explainable AI as future directions. IEEE Internet of Things Journal, 10(5), 4059–4092. https://doi.org/10.1109/JIOT.2022.3203249

[11] Sudheera, K. L. K., Divakaran, D. M., Singh, R. P., & Gurusamy, M. (2021). ADEPT: Detection and identification of correlated attack stages in IoT networks. IEEE Internet of Things Journal, 8(8), 6591–6607. https://doi.org/10.1109/JIOT.2021.3055937

[12] Li, W., Meng, W., & Kwok, L. F. (2022). Surveying trust-based collaborative intrusion detection: State-of-the-art, challenges and future directions. IEEE Communications Surveys & Tutorials, 24(1), 280–305. https://doi.org/10.1109/COMST.2021.3139052

[13] Golomb, T., Mirsky, Y., & Elovici, Y. (2018). CIoTA: Collaborative IoT anomaly detection via blockchain. arXiv. https://doi.org/10.48550/arXiv.1803.03807

[14] Gyamfi, E., & Jurcut, A. (2022). Intrusion detection in Internet of Things systems: A review on design approaches leveraging multi-access edge computing, machine learning, and datasets. Sensors, 22(10), 3744. https://doi.org/10.3390/s22103744

[15] Nguyen, X.-H., Nguyen, X.-D., Huynh, H.-H., & Le, K.-H. (2022). Realguard: A lightweight network intrusion detection system for IoT gateways. Sensors, 22(2), 432. https://doi.org/10.3390/s22020432

[16] Prabhat Kumar, Gupta, G. P., & Tripathi, R. (2021). Design of anomaly-based intrusion detection system using fog computing for IoT network. Automatic Control and Computer Sciences, 55(2), 137–147. https://doi.org/10.3103/S0146411621020085

[17] Man, D., Zeng, F., Yang, W., Yu, M., Lv, J., & Wang, Y. (2021). Intelligent intrusion detection based on federated learning for edge-assisted Internet of Things. Security and Communication Networks, 2021(1), 9361348. https://doi.org/10.1155/2021/9361348
[18] Kairouz, P., McMahan, H. B., Avent, B., Bellet, A., Bennis, M., Bhagoji, A. N., Bonawitz, K., Charles, Z., Cormode, G., Cummings, R., D'Oliveira, R. G. L., Eichner, H., El Rouayheb, S., Evans, D., Gardner, J., Garrett, Z., Gascón, A., Ghazi, B., Gibbons, P. B., … Zhao, S. (2021). Advances and open problems in federated learning. Foundations and Trends® in Machine Learning, 14(1–2), 1–210. https://doi.org/10.1561/2200000083
[19] Blanchard, P., El Mhamdi, E. M., Guerraoui, R., & Stainer, J. (2017). Byzantine-tolerant machine learning. arXiv. https://doi.org/10.48550/arXiv.1703.02757
[20] Yin, D., Chen, Y., Kannan, R., & Bartlett, P. (2018). Byzantine-robust distributed learning: Towards optimal statistical rates. In Proceedings of the 35th International Conference on Machine Learning (pp. 5650–5659). PMLR. https://proceedings.mlr.press/v80/yin18a.html
[21] Li, S., Ngai, E. C.-H., & Voigt, T. (2023). Byzantine-robust aggregation in federated learning empowered industrial IoT. *IEEE Transactions on Industrial Informatics, 19*(2), 1165–1175. https://doi.org/10.1109/TII.2021.3128164
[22] Xie, C., Koyejo, S., & Gupta, I. (2019). Fall of empires: Breaking Byzantine-tolerant SGD by inner product manipulation. arXiv. https://doi.org/10.48550/arXiv.1903.03936
[23] Fang, M., Cao, X., Jia, J., & Gong, N. (2020). Local model poisoning attacks to Byzantine-robust federated learning. In 29th USENIX Security Symposium (USENIX Security 20) (pp. 1605–1622). USENIX Association. https://www.usenix.org/conference/usenixsecurity20/presentation/fang
[24] Bonawitz, K., Ivanov, V., Kreuter, B., Marcedone, A., McMahan, H. B., Patel, S., Ramage, D., Segal, A., & Seth, K. (2017). Practical secure aggregation for privacy-preserving machine learning. In Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security (pp. 1175–1191). Association for Computing Machinery. https://doi.org/10.1145/3133956.3133982
[25] Cui, L., Qu, Y., Xie, G., Zeng, D., Li, R., Shen, S., & Yu, S. (2022). Security and privacy-enhanced federated learning for anomaly detection in IoT infrastructures. IEEE Transactions on Industrial Informatics, 18(5), 3492–3500. https://doi.org/10.1109/TII.2021.3107783
[26] Wei, L., Yang, Y., Wu, J., Long, C., & Li, B. (2022). Trust management for Internet of Things: A comprehensive study. IEEE Internet of Things Journal, 9(10), 7664–7679. https://doi.org/10.1109/JIOT.2021.3139989
[27] Ud Din, I., Bano, A., Awan, K. A., Almogren, A., Altameem, A., & Guizani, M. (2023). LightTrust: Lightweight trust management for edge devices in industrial Internet of Things. IEEE Internet of Things Journal, 10(4), 2776–2783. https://doi.org/10.1109/JIOT.2021.3081422
[28] Campos, E. M., Saura, P. F., González-Vidal, A., Hernández-Ramos, J. L., Bernabé, J. B., Baldini, G., & Skarmeta, A. (2022). Evaluating federated learning for intrusion detection in Internet of Things: Review and challenges. Computer Networks, 203, 108661. https://doi.org/10.1016/j.comnet.2021.108661
[29] Sarhan, M., Lo, W. W., Layeghy, S., & Portmann, M. (2022). HBFL: A hierarchical blockchain-based federated learning framework for collaborative IoT intrusion detection. *Computers & Electrical Engineering, 103*, 108379. https://doi.org/10.1016/j.compeleceng.2022.108379
[30] Marano, S., Matta, V., & Tong, L. (2009). Distributed detection in the presence of Byzantine attacks. IEEE Transactions on Signal Processing, 57(1), 16–29. https://doi.org/10.1109/TSP.2008.2007335
[31] Vempaty, A., Ray, P., & Varshney, P. K. (2014). False discovery rate based distributed detection in the presence of Byzantines. IEEE Transactions on Aerospace and Electronic Systems, 50(3), 1826–1840. https://doi.org/10.1109/TAES.2014.120645